

Nonlinear ILC

Avoiding noise amplification through nonlinear filters

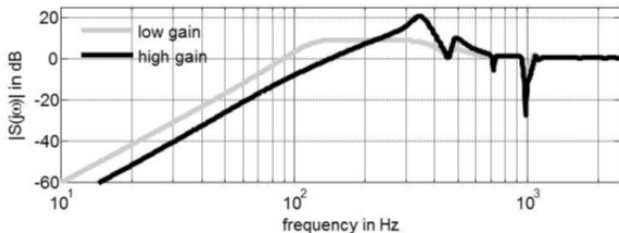
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1. Nonlinear ILC: main idea
2. Non-repeating disturbances in ILC
3. Nonlinear learning and design
4. Simulation example
5. If time permits: convergence/stability for nonlinear ILC/RC

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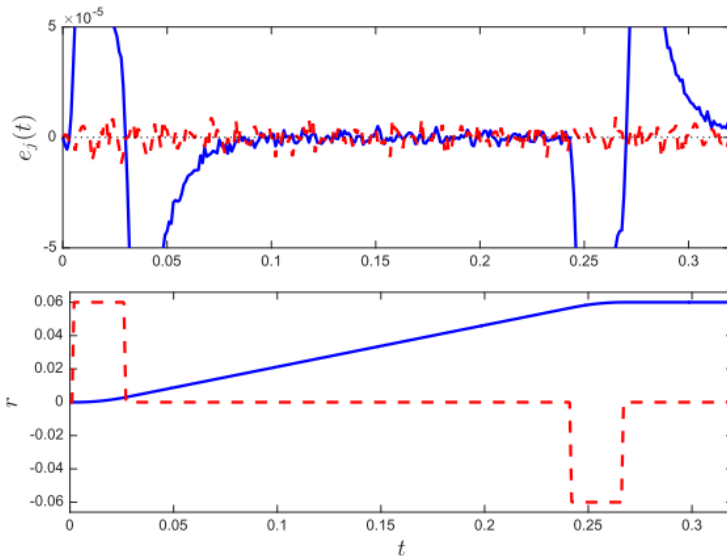
- ▶ Use a nonlinear ILC controller to increase design freedom
- ▶ Inspired by idea of variable-gain feedback control



Which sensitivity for which input?

- ▶ input at 10 Hz
- ▶ input at 350 Hz

And what if the disturbance spectrum is time-varying, e.g., acceleration and constant velocity?



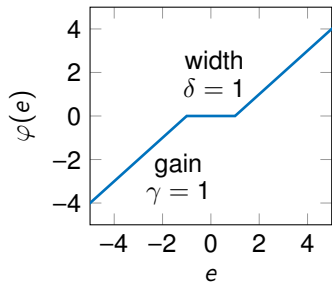
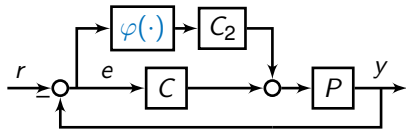
Error and reference signals
wafer stage application

[Oomen and Rojas (Mechatronics 2017)]

Top: error without
feedforward
error with feedforward

Bottom: reference
acceleration

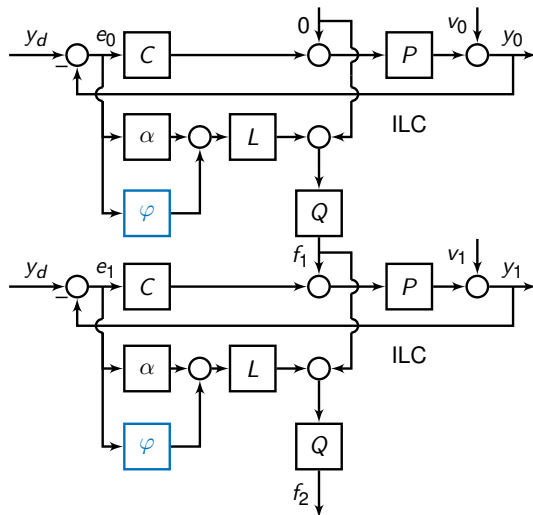
Feedback: low gain
between 0.1 and 0.24, high
gain elsewhere



Extra controller C_2 and deadzone nonlinearity $\varphi(e)$:

$$\varphi(e) = \begin{cases} 0, & \text{if } |e| \leq \delta \\ \left(\gamma - \frac{\gamma\delta}{|e|}\right) e, & \text{if } |e| > \delta. \end{cases}$$

Key idea: apply C_2 only for error signals that exceed δ



But why do this for ILC?

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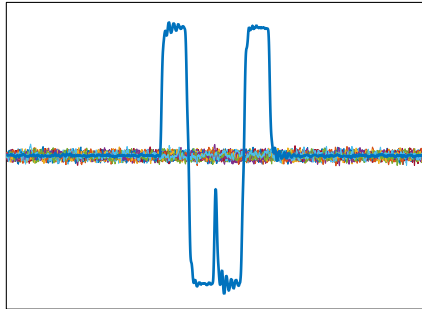
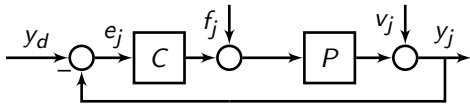
$$e_j = \underbrace{\frac{1}{1+PC}}_S (y_d - v_j) - \underbrace{SP}_J f_j$$

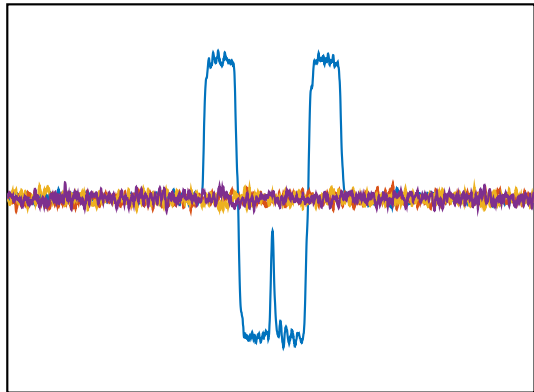
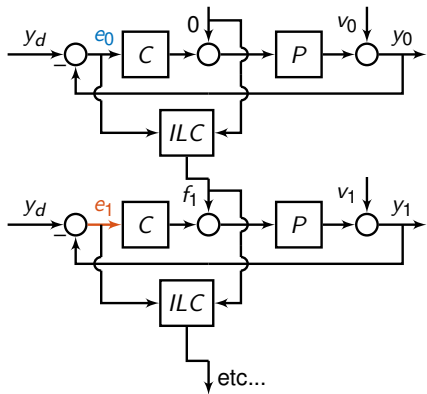
ILC can compensate:

- ▶ everything that repeats (Sy_d)
- ▶ nothing that doesn't repeat (Sv_j)

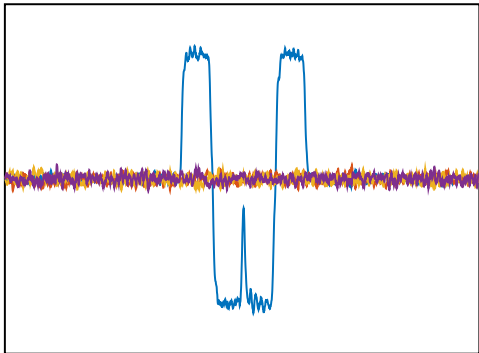
Through n_e experiments with $f_j = 0$:

$$\widehat{Sy_d} = \frac{1}{n_e} \sum_{j=0}^{n_e-1} e_j \quad \text{and} \quad \widehat{Sv_j} = \widehat{Sy_d} - e_j$$

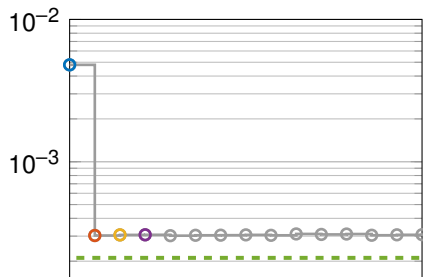




No repeating error after one iteration!

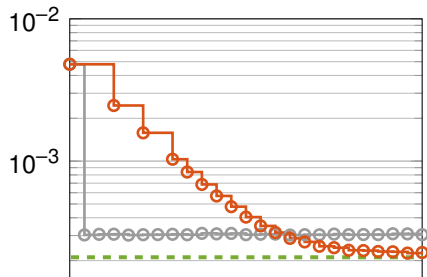


Error 2-norm over iterations:



- error norm is a factor 1.5 above the noise level!

- ▶ Repeating disturbances: fully compensated after a few iterations
- ▶ Iteration-varying disturbances: amplified up to a factor 2
- ▶ Trade-off: limiting the amplification by reducing the learning gain α leads to slow convergence (—)



ILC system: $e_j = S(y_d - v_j) - Jf_j, \quad f_{j+1} = Q(f_j + \alpha L e_j)$

Assume

- ▶ y_d is iteration-invariant
- ▶ v_j is iteration-varying
- ▶ $|Q(1 - \alpha L J)| < 1$ (convergence)

Then at iteration j :

$$f_{j+1} = \underbrace{\sum_{i=1}^j (Q(1 - \alpha L J))^{i-1} Q \alpha L S y_d}_{\text{straightforward part}} - \underbrace{\sum_{n=1}^j (Q(1 - \alpha L J))^{j-n} Q \alpha L S v_j}_{\text{all past noise realizations!}}$$

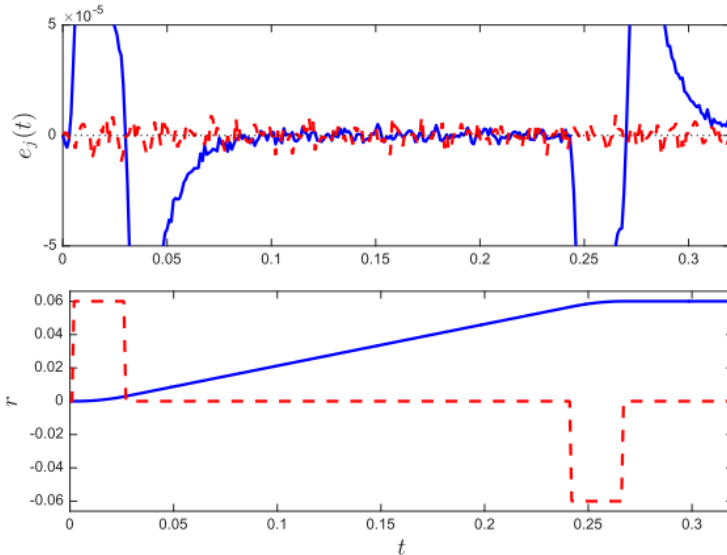
ILC system: $e_j = S(y_d - v_j) - Jf_j, \quad f_{j+1} = Q(f_j + \alpha L e_j)$

Spectrum ϕ_{e_∞} of converged error for $j \rightarrow \infty$ with $L = J^{-1}$ and $Q \neq 1$

$$\phi_{e_\infty} = \left| \frac{1 - Q}{1 - (Q(1 - \alpha))} \right|^2 \phi_{S y_d} + \left(1 + \frac{|\alpha Q|^2}{1 - |Q(1 - \alpha)|^2} \right) \phi_{S v}$$

► Reducing $\alpha \in (0, 1]$ increases $\left| \frac{1 - Q}{1 - (Q(1 - \alpha))} \right|^2$ and increases contribution $\phi_{S y_d}$

Choose $\alpha \approx 0$ for varying disturbances, and $\alpha \approx 1$ for repeating disturbances



Error and reference signals wafer stage application

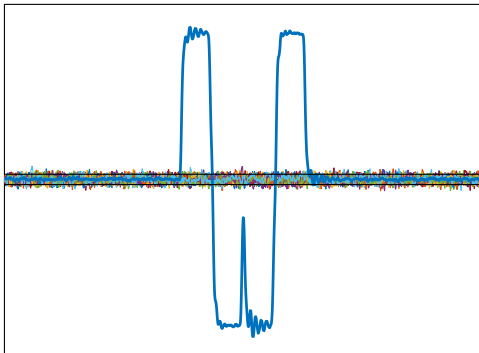
[Oomen and Rojas (Mechatronics 2017)]

Top: **error without
feedforward**
error with feedforward

Bottom: **reference
acceleration**

Learning: $\alpha \approx 0$ between
0.1 and 0.24, $\alpha \approx 1$
elsewhere

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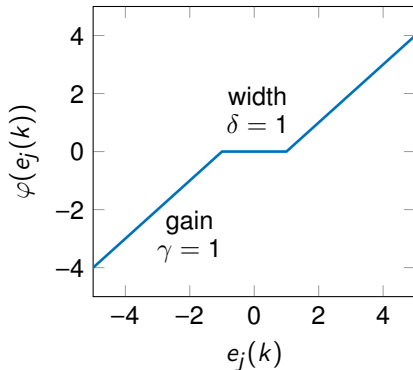
- ▶ Observation: amplitude iteration-varying disturbances is typically smaller
- ▶ Good performance when $\alpha \approx 0$ for varying disturbances and $\alpha \approx 1$ for repeating disturbances
- ▶ Main idea: filter measured error through a deadzone nonlinearity
- ▶ Result: different learning gains based on signal amplitude

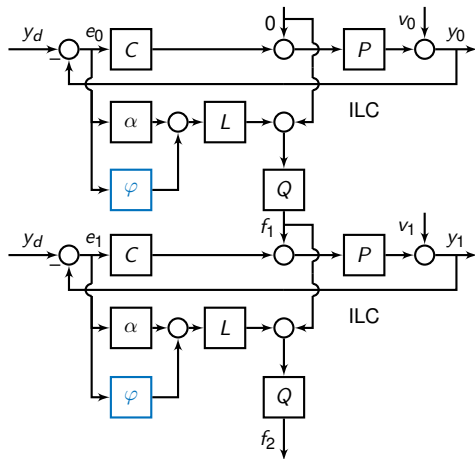
Nonlinear ILC update:

$$f_{j+1} = Q(f_j + \alpha L e_j + L \varphi(e_j))$$

Deadzone nonlinearity $\varphi(e_j)$ is applied to each element k of e_j :

$$\varphi(e_j(k)) = \begin{cases} 0, & \text{if } |e_j(k)| \leq \delta \\ \left(\gamma - \frac{\gamma\delta}{|e_j(k)|}\right) e_j(k), & \text{if } |e_j(k)| > \delta. \end{cases}$$





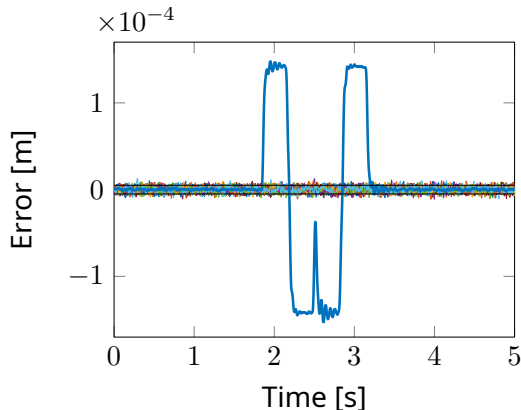
- Resulting update:

$$f_{j+1} = Q(f_j + \alpha L e_j + L \varphi(e_j))$$
- Main idea: choose linear gain α small and nonlinear gain γ large

- $n_e = 20$ experiments with $f_j = 0$

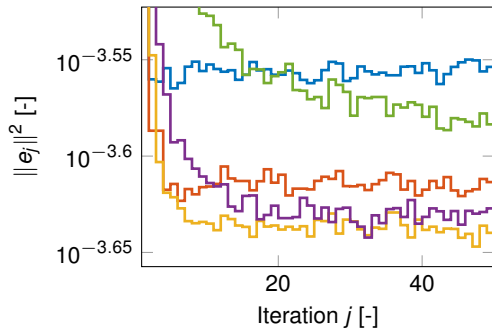
$$\widehat{Sy_d} = \frac{1}{n_e} \sum_{j=0}^{n_e-1} e_j, \quad \widehat{Sv_j} = \widehat{Sy_d} - e_j$$

- Choose δ such that $\varphi(\widehat{Sv_j}) \approx 0$
- Here $\delta = 5 \times 10^{-6}$ (—)



- ▶ The deadzone width δ does not influence convergence/stability
- ▶ However, the performance does depend strongly on δ

We can find the optimal deadzone width automatically!



Deadzone width: 1×10^{-6} ,
 3×10^{-6} , 5×10^{-6} , 7×10^{-6} ,
 10×10^{-6}

Main idea: gradient descent step at each iteration:

$$\delta_{j+1} = \delta_j - \beta \frac{\partial \mathcal{J}(\delta)}{\partial \delta}, \quad \text{with} \quad \mathcal{J}(\delta) = \hat{\mathbf{e}}_{j+1}^T(\delta) \hat{\mathbf{e}}_{j+1}(\delta)$$

- ▶ Error estimate: $\hat{\mathbf{e}}_{j+1}(\delta) = \hat{\mathbf{S}}_{y_d} - \hat{\mathbf{J}} f_{j+1}(\delta)$
- ▶ $\hat{\mathbf{S}} = (1 - \hat{\mathbf{P}}\mathbf{C})^{-1}$ and $\hat{\mathbf{J}}$ are available from ILC design procedure
- ▶ Reference y_d is known
- ▶ Gradient expression: $\frac{\partial \mathcal{J}(\delta)}{\partial \delta} = \left(-2\hat{\mathbf{J}}^T \hat{\mathbf{e}}_{j+1} \right)^T \frac{\partial f_{j+1}}{\partial \delta}$

The derivative $\frac{\partial f_{j+1}}{\partial \delta}$ follows from rewriting the deadzone expression:

$$\frac{\partial f_{j+1}}{\partial \delta} = QL\varphi_2(\mathbf{e}_j, \delta), \quad \text{with } \varphi_2(\mathbf{e}_j(k), \delta) = \begin{cases} 0, & \text{if } |\mathbf{e}_j(k)| \leq \delta \\ -\gamma \text{sign}(\mathbf{e}_j(k)), & \text{if } |\mathbf{e}_j(k)| > \delta. \end{cases}$$

Then, the gradient of criterion $\mathcal{J}$ with respect to δ is given by

$$\frac{\partial \mathcal{J}(\delta)}{\partial \delta} = \left(-2\hat{\mathbf{J}}^\top \hat{\mathbf{e}}_{j+1}(\delta) \right)^\top QL\varphi_2(\mathbf{e}_j, \delta).$$

At each ILC iteration:

1. Estimate $\hat{e}_{j+1}(\delta) = \hat{S}_{y_d} - \hat{J}f_{j+1}(p_\varphi)$ through simulation with $\hat{S}$ and $\hat{J}$
2. Compute derivative $\frac{\partial f_{j+1}}{\partial \delta} = QL\varphi_2(e_j, \delta)$
3. Compute gradient $\frac{\partial \mathcal{J}(p_\varphi)}{\partial p_\varphi} = \left(-2\hat{J}^T \hat{e}_{j+1}\right)^T \frac{\partial f_{j+1}}{\partial \delta}$
4. Update $\delta_{j+1} = \delta_j + \beta \left(2\hat{J}^T \hat{e}_{j+1}(\delta)\right)^T QL\varphi_2(e_j, \delta)$
5. Compute ILC update $f_{j+1} = Q(f_j + \alpha L\hat{J}e_j + L\varphi(\delta_{j+1}, e_j))$

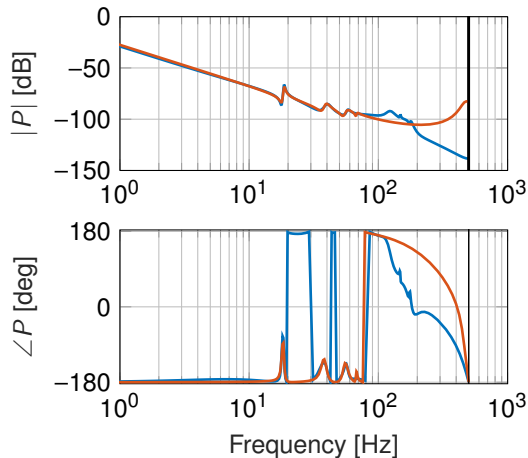
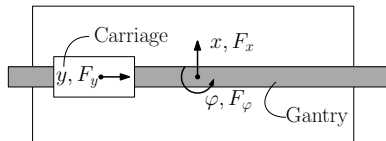
1. Determine deadzone width δ based on experiments without ILC, or use self-optimizing approach:
2. Design $L \approx J^{-1}$ (stable inversion, ZPETC)
3. Design zero-phase lowpass filter Q such that $\|Q(1 - JL)\|_{\mathcal{L}_\infty} < 1$
(linear case with $\alpha = 1$)
4. Choose $\gamma \approx 1$, $\alpha \approx 0$ and $\alpha + \gamma \approx 1$ while satisfying

$$\left\| Q \left(1 - \alpha JL - \frac{\gamma}{2} JL \right) \right\|_{\mathcal{L}_\infty} + \frac{\gamma}{2} \|QJL\|_{\mathcal{L}_\infty} < 1$$

Note similarity of convergence condition to linear case $\implies$ design directly using Bode plots!

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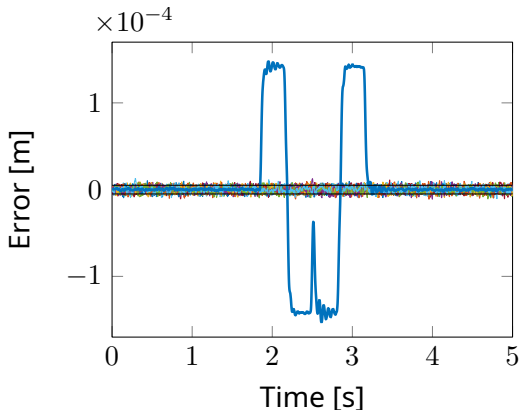
- y -translation of the carriage of a flatbed printer
- System (—) and low-order approximation (—)



- $n_e = 20$ experiments with $f_j = 0$

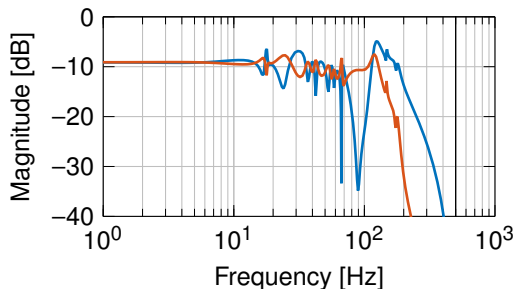
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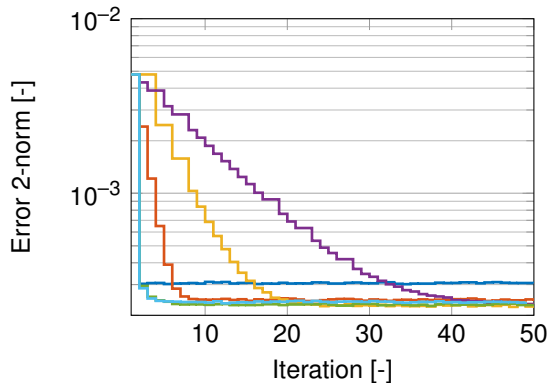
- Choose δ such that $\varphi(\widehat{Sv_j}) \approx 0$
- Here $\delta = 5 \times 10^{-6}$ (—)
- Alternative: self-optimizing approach, only choose initial δ_1



1. Choose δ using 20 experiments without feedforward
2. Design $L \approx J^{-1}$ (stable inversion, ZPETC)
3. Design Q such that $\|Q(1 - JL)\|_{\mathcal{L}_\infty} < 1$ (linear case with $\alpha = 1$)
4. Choose $\gamma \approx 1$, $\alpha \approx 0$ and $\alpha + \gamma \approx 1$ while satisfying

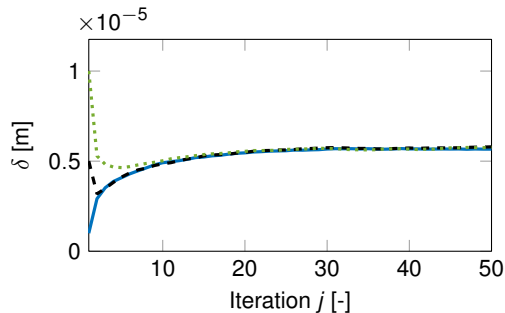
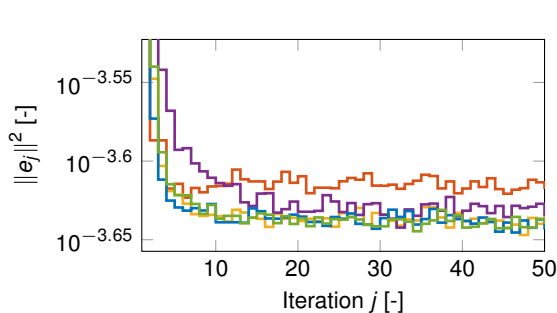
$$\underbrace{\left\| Q \left(1 - \alpha JL - \frac{\gamma}{2} JL \right) \right\|_{\mathcal{L}_\infty}}_{=0.572(\text{for } \alpha=0.3, \gamma=0.7)} + \underbrace{\frac{\gamma}{2} \|QJL\|_{\mathcal{L}_\infty}}_{=0.422} < 1$$





- Nonlinear ILC removes the trade-off between fast convergence and limited amplification of varying disturbances

Linear: $\alpha = 1, 0.5, 0.2, 0.1$ Nonlinear: $\alpha = 0.3, \gamma = 0.7$ and $\alpha = 0.1, \gamma = 0.9$



Deadzone widths:

- Constant: 3×10^{-6} , 5×10^{-6} , and 7×10^{-6}
- Self-optimizing: $\delta_1 = 1 \times 10^{-6}$, $\delta_1 = 5 \times 10^{-6}$ and $\delta_1 = 1 \times 10^{-5}$
- For different initial values, the self-optimizing deadzone width performs as well as the constant, optimal value

- ▶ ILC amplifies non-repeating disturbances
- ▶ Limiting amplification typically leads to slow convergence
- ▶ Solution: use a deadzone in nonlinear ILC to apply different learning gains based on amplitude characteristics
- ▶ Leads to fast convergence and small errors
- ▶ Further reading and related work:
 - ▶ Also works for lifted ILC [Aarnoudse, Pavlov and Oomen (Automatica 2025)] and repetitive control
 - ▶ Method and design procedure in [Aarnoudse, Pavlov and Oomen (CEP 2024, CDC 2025)]

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Theorem: Monotonic convergence of nonlinear frequency-domain ILC

The sequence of iterates $\{e_j\}$ is monotonically convergent in the 2-norm for $v_j = 0 \forall j$ if

$$\left\| Q \left(1 - \alpha JL - \frac{\gamma}{2} JL \right) \right\|_{\mathcal{L}_\infty} + \frac{\gamma}{2} \|QJL\|_{\mathcal{L}_\infty} < 1$$

- ▶ $\mathcal{L}_\infty$ -norms can be evaluated using measured frequency-response data
- ▶ Intuitive tuning of robustness filter Q using Bode diagrams

1. $\varphi(\mathbf{e}_j)$ satisfies an incremental sector condition. For any scalars a, b

$$0 \leq \frac{\varphi(a) - \varphi(b)}{a - b} \leq \gamma$$

2. Loop transformation to a symmetric sector condition. $\tilde{\varphi} = \varphi(\mathbf{e}_j) - \frac{\gamma}{2}\mathbf{e}_j$ and

$$-\frac{\gamma}{2} \leq \frac{\tilde{\varphi}(a) - \tilde{\varphi}(b)}{a - b} \leq \frac{\gamma}{2}$$

3. Ensure that there exists $\rho < 1$ for which

$$\|\mathcal{F}(\mathbf{e}_1) - \mathcal{F}(\mathbf{e}_2)\|_2 \leq \rho \|\mathbf{e}_1 - \mathbf{e}_2\|_2$$

- ▶ [Aarnoudse, Pavlov and Oomen (CEP 2024)] Leontine Aarnoudse, Alexey Pavlov and Tom Oomen, "A design framework for nonlinear iterative learning control and repetitive control: Applied to three mechatronic case studies," *Control Engineering Practice*, vol. 149, pp. 105976, 2024. DOI:10.1016/j.conengprac.2024.105976
- ▶ [Aarnoudse, Pavlov and Oomen (Automatica 2025)] Leontine Aarnoudse, Alexey Pavlov and Tom Oomen, "Nonlinear iterative learning control for discriminating between disturbances," *Automatica*, vol. 171, pp. 111902, 2025. DOI:10.1016/j.automatica.2024.111902
- ▶ [Aarnoudse, Pavlov and Oomen (CDC 2025)] Leontine Aarnoudse, Alexey Pavlov and Tom Oomen, "Self-optimization of nonlinear iterative learning control and repetitive control," *Proceedings of the 63rd IEEE Conference on Decision and Control*, pp. 765-770, Rio de Janeiro, Brazil, 2025
- ▶ [Oomen and Rojas (Mechatronics 2017)] Tom Oomen and Cristian Rojas, "Sparse iterative learning control with application to a wafer stage," *Mechatronics*, vol. 47, pp. 134-147, 2018. DOI:10.1016/j.mechatronics.2017.09.004